

$$S_{14} = S_{15} = \dots = S_{1n-1} = 0$$

$$S_{24} = S_{25} = \dots = S_{2n-1} = 0$$

$$S_{1j} = 0$$

$$S_{2j} = 0$$

$$j = 4, \dots, n-1$$

$$0 = X_{1j} + X_{2j+1} - X_{1j+1} - X_{2j}$$

$$0 = X_{2j} + X_{3j+1} - X_{2j+1} - X_{3j}$$

$$\oplus \Rightarrow X_{3j+1} = X_{1j+1} - X_{1j} + X_{3j}$$

$$\begin{cases} X_{35} = X_{15} - X_{14} + X_{34} = X_{15} - X_{14} \\ X_{36} = X_{16} - X_{15} + X_{35} = X_{16} - X_{14} \\ \vdots \\ X_{3n-1} = X_{1n-1} - X_{14} \\ X_{3n} = -X_{14} \end{cases}$$

$$X_{25} = X_{35} + X_{24} - X_{34} = X_{35} + X_{24}$$

$$X_{26} = X_{36} + X_{25} - X_{35} = X_{36} + X_{24}$$

$$\vdots$$

$$X_{2n-1} = X_{3n-1} + X_{24}$$

$$X_{2n} = X_{3n} + X_{24}$$

$$\Rightarrow X_{2j} = X_{3j} + X_{24}$$

$$\begin{cases} X_{35} = X_{15} - X_{14} \\ X_{36} = X_{16} - X_{14} \\ \vdots \\ X_{3n-1} = X_{1n-1} - X_{14} \\ X_{3n} = -X_{14} \\ X_{2j} = X_{3j} + X_{24} \quad (j = 5, \dots, n) \end{cases}$$

thing that can be independent:

$$\begin{array}{ccc} 46 & 47 & \dots & 4n \\ 57 & \dots & 5n & \oplus & 13 \\ & & \vdots & & \\ & & n-2n & & \end{array}$$

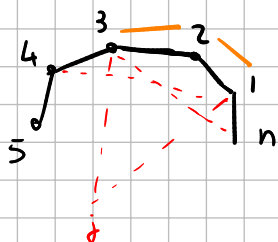
then we pick one each from the pair $(3j, 1j)$ $j = 5, \dots, n-1$

pick one X_{2j} to be independent.

exclusively

X_{14} and X_{3n}

→ each term that has only elements from this list survives

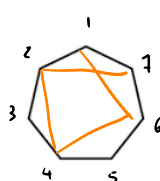
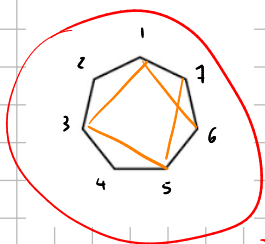
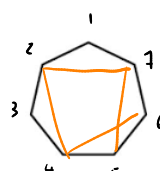
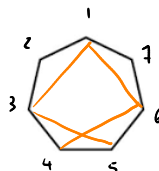
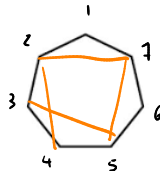
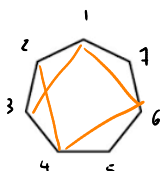
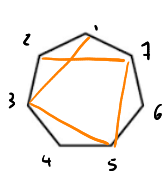


if it has one of:

- $3n$
- 14
- both $1j$ and $3j$ $j = 5, \dots, n-1$
- has 2 or more X_{2j} ($j = 4, \dots, n$)

the term survives

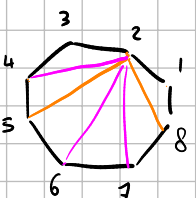
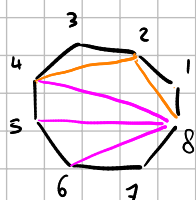
$n = 7$



look at (17)(76)

7 only has 1 connection
and (17), (76) are not
part of a 4-gon
so doesn't survive.

can be a triangulation!



possibly wrong (should be none)

if ② connects to at most 1 other vertex $i \in \{4, \dots, n\}$
then the edges (12), (23) must be part of a 4-gon

$K=0$

$$\begin{cases} C_{1,k+2} = C_{1,k+3} = \dots = C_{1,n-1} = 0 \\ C_{2,k+2} = C_{2,k+3} = \dots = C_{2,n-1} = 0 \\ \vdots \\ C_{k,k+2} = C_{k,k+3} = \dots = C_{k,n-1} = 0 \end{cases}$$

$K=1, \dots, n-3$

$\Rightarrow C_{1j} = C_{2j} = \dots = C_{kj} = 0 \quad j = k+2, \dots, n-1$

$C_{ij} = X_{i,i} + X_{i,i+j} - X_{i,j+1} - X_{i+j,j}$

$X_{2j+1} + X_{1j} - X_{1,j+1} - X_{2j} = 0$

$X_{3j+1} + X_{2j} - X_{2,j+1} - X_{3j} = 0$

$\vdots$

$X_{k+1,j+1} + X_{kj} - X_{k,j+1} - X_{k+1,j} = 0$

$i = 1, 2, \dots, k$

add all equations:

$X_{k+1,j+1} = X_{k+1,j} + X_{1,j+1} - X_{1,j}$

$X_{k+1,k+3} = X_{k+1,k+2} + X_{1,k+3} - X_{1,k+2}$

$j = k+2$

$X_{k+1,k+4} = X_{k+1,k+3} + X_{1,k+4} - X_{1,k+3} = X_{1,k+4} - X_{1,k+2}$

$j = k+3$

$$\rightarrow \left\{ \begin{array}{l} X_{k+1, k+3} = X_{1, k+3} - X_{1, k+2} \\ X_{k+1, k+4} = X_{1, k+4} - X_{1, k+2} \\ X_{k+1, k+5} = X_{1, k+5} - X_{1, k+2} \\ \vdots \\ X_{k+1, n-1} = X_{1, n-1} - X_{1, k+2} \\ X_{k+1, n} = -X_{1, k+2} \end{array} \right. \quad \left\{ \begin{array}{l} X_{2j} = X_{k+1, j} + X_{2, k+2} \\ X_{3j} = X_{k+1, j} + X_{3, k+2} \\ \vdots \\ X_{kj} = X_{k+1, j} + X_{n, k+2} \end{array} \right. \quad j = k+3, \dots, n$$

if a term has at most one element from each pair (X_{1j}, X_{k+1j}) with $j \in \{k+3, \dots, n-1\}$ and does not have

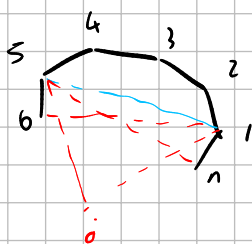
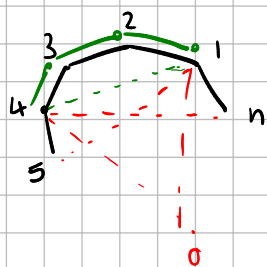
any $X_{2j}, X_{3j}, \dots, X_{kj}$ with $j \in \{k+2, \dots, n\}$

and does not have $X_{k+1, n}$ or $X_{1, k+2}$

then the term has coefficient zero. $\rightarrow$ proved by sending the excluded things to ∞

$(k=3)$ if we exclude $X_{4n}, X_{15}, X_{2j}, X_{3j}$ with $j \geq 5$, and keep at most one from each (X_{1j}, X_{4j}) then the term vanishes

$\rightarrow j \in \{6, \dots, n-1\}$



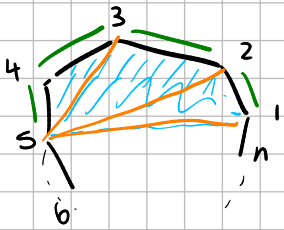
• if vertices $2, 3$ only connect to vertices inside $(1, 2, 3, 4)$

then the edges $(12), (23), (34)$ must be part of a 5-gon.

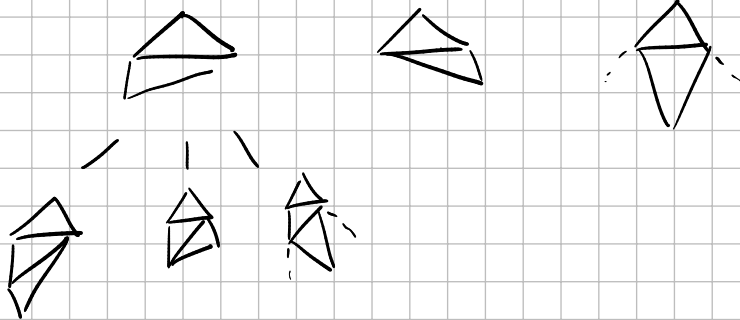
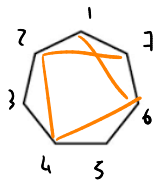
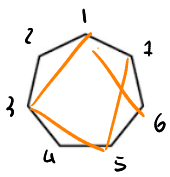
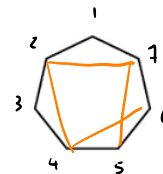
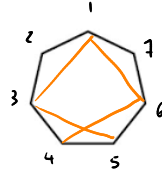
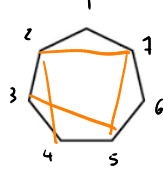
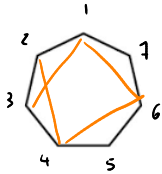
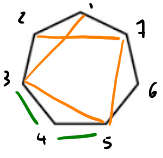
$\rightarrow$ equivalently, they don't cross chord $(1, k+1)$

$\rightarrow$ if vertices $2, 3, 4, \dots, k$ only connect to vertices inside $(1, 2, 3, 4, \dots, k+1)$

then the edges $(12), (23), \dots, (k, k+1)$ must be part of a $(k+2)$ -gon.



test $n=7$



$n=8$

